

Model Concepts

1 Train Test Split

Dividing dataset into two parts:

- **Training set** — model learns from this
- **Test set** — model is evaluated on this

Why we need it?

If you test on the same data you trained on — model already saw those answers. Like giving a student the exact same exam questions they studied. Score will be high but meaningless.

Test set simulates **real world unseen data**.

Standard Split

80% Training — model learns

20% Testing — model is evaluated

Sometimes:

70% Training

15% Validation — tune the model

15% Testing — final evaluation

Validation Set vs Test Set

Validation set — used during training to tune model. Model indirectly sees this.

Test set — used only once at the very end. Model never sees this during training. This gives true performance.

Key Parameter — `random_state`

Always set `random_state=42` or any fixed number. This ensures same split every time you run the code. Without it — different runs give different splits and different results.

Stratified Split

For classification with imbalanced data — use `stratify=y`. This ensures train and test sets have same class distribution.

Dataset: 80% no disease, 20% disease

Without stratify: Train might have 85% no disease — biased

With stratify: Train has exactly 80% no disease — balanced

2 Cross Validation

Problem with Single Train Test Split

One split can be lucky or unlucky. Maybe test set has easy examples — model looks better than it is. Maybe test set has hard examples — model looks worse.

Single split gives unreliable estimate of model performance.

What is Cross Validation?

Evaluate model on **multiple different splits** and average the results. Much more reliable.

K-Fold Cross Validation — Most Common

Split data into K equal parts called folds.

Example — 5 Fold Cross Validation:

Fold 1: [Test] [Train] [Train] [Train] [Train]

Fold 2: [Train] [Test] [Train] [Train] [Train]

Fold 3: [Train] [Train] [Test] [Train] [Train]

Fold 4: [Train] [Train] [Train] [Test] [Train]

Fold 5: [Train] [Train] [Train] [Train] [Test]

Final score = average of all 5 scores

Model is trained 5 times. Each time different fold is test set. Final score is average — much more reliable.

Stratified K-Fold

Same as K-Fold but maintains class distribution in each fold. Always use this for classification problems.

When to use Cross Validation

- Small datasets — cannot afford to lose data for validation
 - When you want reliable performance estimate
 - When comparing multiple models
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3 Overfitting & Underfitting

Underfitting

Model is **too simple** to learn patterns in data.

Training accuracy — low

Test accuracy — low

Like a student who barely studied. Fails both practice and real exam.

Causes:

- Model too simple — linear model for complex data
- Too few features
- Too little training

Solutions:

- Use more complex model
 - Add more features
 - Train longer
-

Overfitting

Model **memorizes training data** including noise. Cannot generalize to new data.

Training accuracy — very high (95%+)

Test accuracy — low (60-70%)

Big gap between train and test accuracy

Like a student who memorized answers word by word. Passes practice exam perfectly but fails real exam with slightly different questions.

Causes:

- Model too complex
- Too many features
- Too little training data
- Training too long

Solutions:

- Regularization — Ridge, Lasso, Dropout
- Cross Validation
- More training data
- Feature selection — remove irrelevant features

- Early stopping — stop training when validation loss increases

Good Fit

Training accuracy — high

Test accuracy — similar to training accuracy

Small gap between both

How to Detect

Train accuracy high, Test accuracy high → Good fit ✓

Train accuracy low, Test accuracy low → Underfitting ✗

Train accuracy high, Test accuracy low → Overfitting ✗

Bias Variance Tradeoff

Bias

Error from wrong assumptions in the model. High bias means model is too simple and misses important patterns.

High bias → Underfitting

Variance

Error from sensitivity to small changes in training data. High variance means model changes a lot with different training data.

High variance → Overfitting

The Tradeoff

Simple model → High Bias, Low Variance → Underfitting

Complex model → Low Bias, High Variance → Overfitting

Goal → Low Bias, Low Variance → Good fit

You cannot minimize both at the same time — there is always a tradeoff. Goal is to find the right balance.

Real Example

Linear model on complex data:

High bias — misses the curve

Low variance — consistent but consistently wrong

Deep decision tree:

Low bias — fits every point perfectly

High variance — changes completely with new data

5 Hyperparameter Tuning

What are Hyperparameters?

Parameters you set **before** training. Model does not learn these — you choose them.

KNN → n_neighbors (K value)

Decision Tree → max_depth, min_samples_split

Random Forest → n_estimators, max_depth

XGBoost → learning_rate, n_estimators, max_depth

SVM → C, kernel, gamma

Parameters vs Hyperparameters

Parameters

Learned by model during training

Weights and coefficients

Cannot be set manually

Hyperparameters

Set by you before training

K value, depth, learning rate

Must be set manually

Techniques for Tuning

1 — Manual Tuning: Try different values yourself. Slow and inefficient.

2 — Grid Search: Try every possible combination of hyperparameters you define. Exhaustive but slow.

Try all combinations of:

K = [3, 5, 7, 9]

metric = [euclidean, manhattan]

Total combinations = $4 \times 2 = 8$ — tries all 8

3 — Random Search: Try random combinations instead of all. Faster than grid search. Often finds good results.

4 — Bayesian Optimization: Smarter search — uses previous results to decide next combination to try. Most efficient.

Which to use?

- Small parameter space → Grid Search
 - Large parameter space → Random Search
 - Production projects → Bayesian Optimization
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Pipelines in Sklearn

What is a Pipeline?

A pipeline chains multiple steps together into one object. Instead of applying each step separately — pipeline does everything in order automatically.

Why use Pipelines?

Without pipeline — messy and error prone:

Step 1 — Scale training data

Step 2 — Train model on scaled data

Step 3 — Scale test data separately

Step 4 — Predict on scaled test data

Risk — easy to forget to scale test data

— data leakage possible

With pipeline — clean and safe:

Pipeline automatically:

→ Scales data

→ Trains model

→ Scales new data before prediction

No risk of forgetting steps

No data leakage

Data Leakage — Critical Concept

When information from test set accidentally influences training. This gives falsely optimistic results. Pipeline prevents this automatically.

Example of leakage without pipeline:

Scale entire dataset first — including test set

Then split into train and test

Problem — scaler learned from test set too

- test set influenced the scaling
- results are not realistic

Correct order with pipeline:

Split first

Pipeline fits scaler only on training data

Applies same scaler to test data

No leakage

7 ROC Curve & AUC Score

What is ROC Curve?

ROC — Receiver Operating Characteristic curve. It shows model performance at **all possible thresholds**.

By default classification threshold is 0.5 — probability above 0.5 means positive class. But you can change this threshold.

ROC curve plots:

- **True Positive Rate (Recall)** on Y axis
- **False Positive Rate** on X axis

At different thresholds — both rates change. ROC curve shows all these points.

AUC — Area Under Curve

AUC measures the area under the ROC curve.

AUC = 1.0 → Perfect model

AUC = 0.9 → Excellent

AUC = 0.8 → Good

AUC = 0.7 → Fair

AUC = 0.5 → Random guessing — useless model

AUC < 0.5 → Worse than random

Why AUC is Better than Accuracy

For imbalanced data — accuracy is misleading. AUC measures how well model separates classes regardless of threshold or class balance.

Always report AUC for medical datasets like heart disease.

Interview Answers — Practice These

What is the difference between train, validation and test set?

"Training set is used to train the model. Validation set is used during training to tune hyperparameters and monitor performance. Test set is used only once at the very end to give the true unbiased performance estimate. The model never sees test data during training."

What is Cross Validation and why use it?

"Cross validation evaluates model on multiple different splits of data and averages the results. K-Fold cross validation splits data into K folds, trains on K-1 folds and tests on remaining fold, repeating K times. This gives a much more reliable performance estimate than a single train test split, especially for small datasets."

What is the difference between overfitting and underfitting?

"Underfitting happens when model is too simple — high training and test error both. Overfitting happens when model memorizes training data — very high training accuracy but low test accuracy. We detect overfitting by comparing training and test accuracy. We fix it using regularization, cross validation, feature selection or more training data."

What is Bias Variance Tradeoff?

"Bias is error from oversimplified assumptions — causes underfitting. Variance is sensitivity to small changes in data — causes overfitting. Simple models have high bias and low variance. Complex models have low bias and high variance. The goal is finding the right balance where both bias and variance are acceptably low."

What is a Pipeline in sklearn?

"A pipeline chains preprocessing and model training steps into one object. It prevents data leakage by ensuring the scaler is fit only on training data and applied consistently to test data. It also makes code cleaner, reproducible and easier to deploy."

What is AUC ROC?

"ROC curve plots true positive rate against false positive rate at all possible thresholds. AUC is the area under this curve — ranges from 0.5 for random model to 1.0 for perfect model. AUC is better than accuracy for imbalanced datasets like medical data because it measures how well the model separates classes regardless of threshold."